(§) This is an optimistic bound, assuming the adjacency matrix can be resized dynamically. Otherwise, a truer bound is O(|V|2).

Keeping this in mind, we'll now provide a definition, one that will be handy later in this section:

A graph G=(V,E) is said to be *sparse* if |E|=O(|V|); G is said to be *dense* when |E|=O(|V|2).

In other words, sparse graphs have a number of edges comparable to the number of vertices, which are therefore loosely connected to each other, while in dense graphs each vertex is connected to most of the other vertices.

Table 14.1 summarizes the PROs and CONs of the two different representation for graphs. In a few words, the adjacency list representation is better for sparse graphs because it requires a lot less memory, and because for sparse graphs |E|≈|V|, and thus most operations can be performed efficiently.

For dense graphs, conversely, since the number of edges is close to the maximum possible, the adjacency matrix representation is more compact and efficient; moreover, this representation allows implementing more efficiently some algorithms on graphs, like the search of connected components or transitive closure.

In general, when no assumption can be made on the graph and unless is otherwise required by the context, the adjacency list representation is preferred because it is more flexible and it support adding new vertices more easily.